

# Applied Fluid Mechanics

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## Engineering Technology

May, 2026

**Short Title:** Applied Fluid Mechanics  
**Faculty** Engineering  
**Department** Engineering Technology  
**Cluster** Engineering  
**Author** GGALLAGHER  
**Credits** 5

**Level:** Advanced

## Description of Module / Aims

This module builds on the fundamentals of fluids established in the Fluid Mechanics module in year 3 of this programme. The approach is to look at real fluids in real applications. It further develops the principles underlying the subject and demonstrates how these are used for the design of hydraulic components and systems.

## Programmes

|           |                                                                      |           |
|-----------|----------------------------------------------------------------------|-----------|
| FP0W-0005 | BEng (Hons) in Mechanical and Manufacturing Engineering (WD_EMSEN_B) | 4 / 8 / M |
| FP0W-0005 | BSc (Hons) in Physics for Modern Technology (WD_KPHTe_B)             | 4 / 8 / E |

## Indicative Content

- Review of basic fluid properties. Fluid statics. Bernoulli and continuity equations.
- Behaviour of real fluids. Flow in bounded systems
- Flow in pipelines. Friction effects. General energy equation. Pump sizing and selection
- External flow: boundary layer, drag and lift
- Dimensional analysis and similarity
- Analysis of fluid flow in rotodynamic machinery such as: centrifugal pump, Francis turbine and Pelton wheel

## Learning Outcomes

*On successful completion of this module, a student will be able to:*

1. Design simple fluids systems accounting for the various energy losses the fluid experience and select a suitable pump.
2. Explain the origin of lift and drag in a variety of external flow scenarios and evaluate consequent drag and lift forces.
3. Apply dimensional analysis and similarity techniques.
4. Analyse simple fluid flows and determine important fluid parameters in such flows.
5. Describe a range of turbomachines and analyse fluid flow in same.

## Learning and Teaching Methods

- Learning and teaching methods and strategies:
  1. Lectures
  2. Assignments
  3. Laboratories

## Assessment Methods

|                            | <b>Weighing</b> | <b>Outcomes Assessed</b> |
|----------------------------|-----------------|--------------------------|
| Final Written Examination  | 70%             | 1,2,3,4,5                |
| Continuous Assessment      | 30%             |                          |
| <i>In-Class Assessment</i> | 15%             | 1,4                      |
| <i>Project</i>             | 15%             | 1,4                      |

## Assessment Criteria

## Learning Modes

| <b>Learning Mode</b> | <b>F/T Hours</b> | <b>P/T Hours</b> |
|----------------------|------------------|------------------|
| Lecture              | 48               |                  |
| Independent Learning | 87               |                  |

## Supplementary Material

- Cengel, Y. and J.M. Cimbala. *\_Fluid Mechanics: Fundamentals and Applications\_*. USA: McGraw-Hill, 2013.
- Douglas, J.F., J.M. Gasiorek, J.A. Swaffield and L.B. Jack. *\_Fluid Mechanics\_*. UK: Prentice Hall, 2011.
- Mott, R.L. *\_Applied Fluid Mechanics\_*. USA: Prentice Hall, 2006.
- Potter, M., D.C. Wiggert and B.H. Ramadan. *\_Mechanics of Fluids\_*. USA: Cengage, 2011.